

SpaRe: Spatial Representation Reward for Vision-Language-Action Reinforcement Learning

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Abstract

*Vision-language-action (VLA) policies trained with reinforcement learning can achieve high robot-manipulation performance, but they often suffer from sparse rewards and limited robustness to changing spatial configurations. This report introduces **Spatial Representation Reward (SpaRe)**, a reward-shaping framework that augments a VLA policy with an auxiliary spatial latent-alignment signal from a pretrained spatial encoder. In a resource-constrained offline-RL implementation on SimpleVLA-RL and LIBERO-10, SpaRe trains from an official supervised OpenVLA-OFT checkpoint using full expert demonstrations plus 409 real failed policy rollouts, while updating only a small subset of policy layers. Compared with the official SFT baseline, the R_1 -only variant improves success rate from 16.13% to 17.20%, and combining R_1 with λR_2 further improves success rate to 18.40%. Although the gains are modest, they indicate that spatial representation reward provides a measurable positive signal under limited compute. Code is publicly available at <https://github.com/Leo-Haochen-Liu/spare-lite-libero10-offline-rl>.*

1. Introduction

Vision-language-action (VLA) models map visual observations and natural-language instructions directly into robot actions, offering a path toward general robot policies that can follow language-specified goals. Recent systems such as OpenVLA and OpenVLA-OFT show that large pretrained models provide effective initialization for manipulation, while reinforcement learning can further improve task success.

However, reinforcement learning for VLA manipulation still faces two central challenges. First, robot rewards are often sparse, leaving many intermediate states weakly supervised. Second, manipulation depends heavily on spatial configuration: a policy must understand object locations, relations, and scene changes, not only the language goal.

This motivates the central research question: can spatial representations provide an additional training signal for VLA reinforcement learning? If so, policies could learn not only from final task success, but also from denser information about spatial relationships, object layouts, and scene geometry.

2. Proposed Method

This work proposes **SpaRe**, a reward-shaping framework that incorporates spatial representation feedback into VLA reinforcement learning. The action policy remains responsible for environment interaction, while an auxiliary reward measures alignment between the policy’s internal representation and a spatially grounded reference representation.

2.1 Spatial Representation Reward Formulation

Let z_t^O denote the policy-side latent representation at time step t , and let z_t^S denote the reference spatial latent extracted from the same observation by a pretrained spatial encoder. The auxiliary spatial reward is defined using cosine similarity:

$$R_{2,t} = \beta_{\text{sp}} \text{sim}(z_t^O, z_t^S), \quad \text{sim}(z_t^O, z_t^S) = \frac{z_t^O{}^\top z_t^S}{\|z_t^O\|_2 \|z_t^S\|_2}. \quad (1)$$

The shaped reward is:

$$R_t = R_{1,t} + \lambda R_{2,t}. \quad (2)$$

Here R_1 is the primary task/data reward, R_2 is the auxiliary spatial-alignment reward, β_{sp} scales the cosine-similarity reward, and λ controls the strength of the spatial reward in the final shaped reward. When $\lambda = 0$, training uses R_1 only. When $\lambda > 0$, training uses a combination of R_1 and R_2 . The objective is not to replace the original reward, but to augment it with denser spatial supervision.

2.2 Rationale for Spatial Reward Shaping

SpaRe is motivated by the spatial structure of robot manipulation. Many manipulation failures arise from insufficient spatial grounding: the policy may not represent object relations, scene layout, or geometric changes in a form that supports reliable control. A pretrained spatial encoder provides a reference representation of these scene properties. By rewarding agreement between the policy latent and the spatial reference latent, SpaRe supplies an intermediate training signal before final task success or failure.

This design is deliberately conservative: R_2 biases learning toward spatially consistent internal representations, while R_1 keeps optimization grounded in the original manipulation objective.

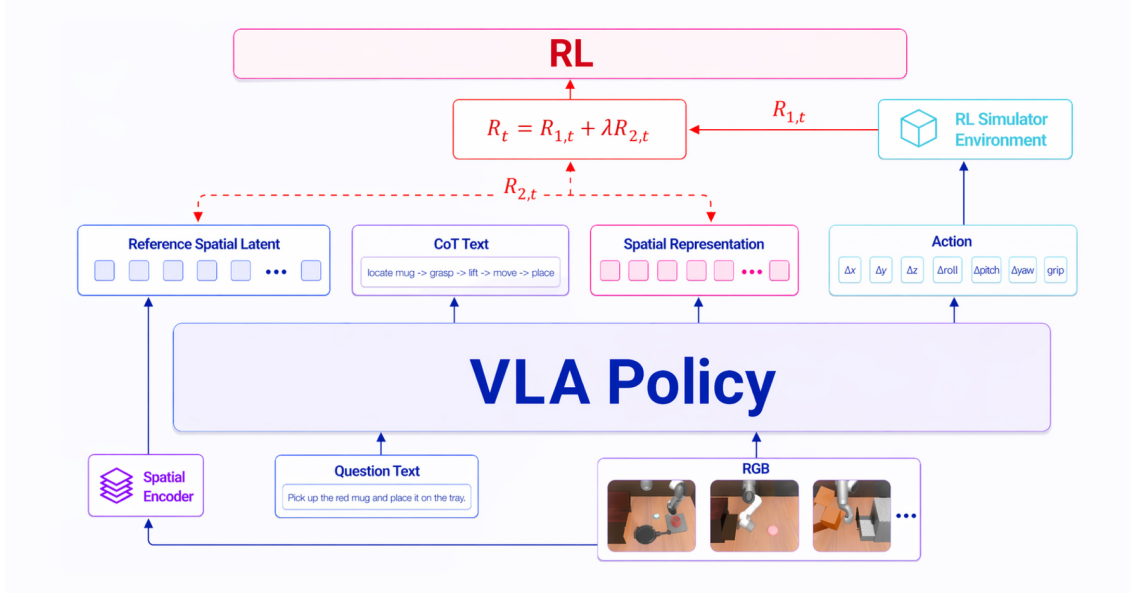


Figure 1: SpaRe training and evaluation pipeline. Expert demonstrations and failed SFT rollouts are converted into labeled offline transitions, optimized with R_1 or $R_1 + \lambda R_2$, exported into full OpenVLA-OFT checkpoints, and evaluated with LIBERO rollouts.

2.3 Offline Optimization Objective

Because full online RL is computationally expensive, the final implementation instantiates SpaRe in an offline RL setting. Given a fixed transition dataset $\mathcal{D}$, where each transition contains a state s , action a , reward r , and next state s' , the method trains an IQL-style actor-critic objective with CQL-style regularization. The learned components are a Q-function $Q_\psi(s, a)$, which estimates the value of taking action a in state s ; a value function $V_\eta(s)$, which estimates the value of state s ; and a policy $\pi_\theta(a|s)$, which parameterizes the action distribution:

$$\mathcal{L}_Q = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(Q_\psi(s, a) - (r + \gamma V_\eta(s')))^2 \right], \quad (3)$$

$$\mathcal{L}_V = \mathbb{E}_{(s,a) \sim \mathcal{D}} [L_2^\tau(Q_\psi(s, a) - V_\eta(s))], \quad (4)$$

$$\mathcal{L}_\pi = -\mathbb{E}_{(s,a) \sim \mathcal{D}} [\exp(\beta_{\text{IQL}} A(s, a)) \log \pi_\theta(a|s)], \quad (5)$$

$$A(s, a) = Q_\psi(s, a) - V_\eta(s).$$

Here $\mathcal{L}_Q$ is the Q-function regression loss, $\mathcal{L}_V$ is the value-function expectile loss, and $\mathcal{L}_\pi$ is the policy loss. The advantage $A(s, a)$ measures how much better action a is than the average value of state s . The discount factor is γ , the expectile parameter in $L_2^\tau(\cdot)$ is τ , and β_{IQL} controls the strength of advantage weighting in the policy update. The expectile value loss biases the value estimate toward high-value actions, while the policy loss weights action imitation by estimated advantage. The objective also uses CQL-style regularization to penalize overestimated Q-values on out-of-distribution actions, which is important in offline RL because the policy cannot collect corrective rollouts during training.

2.4 Overall Pipeline

The overall pipeline first prepares offline transition data, then trains the policy with the shaped reward objective, and finally exports the adapted checkpoint for online LIBERO evaluation. Figure 1 summarizes the interaction among expert demonstrations, failed policy rollouts, the VLA policy, and the spatial reward branch.

3. Related Work

3.1 Vision-Language-Action Policies

Robot foundation models such as RT-1 and RT-2 demonstrated that transformer-based policies can scale with diverse robot data and language-conditioned control tasks [1,2]. More recent open-source VLA models, including Octo and OpenVLA, facilitate reproduction and adaptation [3,4]. OpenVLA-OFT is particularly relevant because it improves the fine-tuning recipe for OpenVLA and reports high LIBERO performance [5].

SpaRe uses an existing VLA checkpoint rather than proposing a new VLA backbone. The difference is that SpaRe studies how to improve the policy through an additional reward signal.

3.2 Reinforcement Learning for VLA Policies

Reinforcement learning for VLA policies is an emerging direction because supervised fine-tuning depends on expensive demonstrations and may generalize poorly under distribution shift. SimpleVLA-RL is the closest method for this setting: it applies online RL to OpenVLA-OFT and reports state-of-the-art LIBERO results [6]. Accordingly, it is used as the primary directly related VLA-RL baseline in this report.

SpaRe is similar to SimpleVLA-RL in that both aim to improve a supervised VLA policy with reinforcement learning and both evaluate through LIBERO rollouts. The differ-

ence is that SimpleVLA-RL focuses on scaling online RL, while this study examines a resource-constrained offline setting. SpaRe also adds a spatial representation reward, whereas SimpleVLA-RL mainly optimizes the task reward produced by the rollout environment.

3.3 Spatial Representations for Manipulation

Spatial representations are important in robot manipulation because many decisions depend on object relations, layout, geometry, and relative motion. Existing spatial manipulation methods include PerAct, RVT, and Act3D [7,8,9]. SpatialVLA brings this idea into VLA models through spatially aware representations and action grids [10].

SpaRe shares the same motivation but uses spatial representations differently: instead of changing the policy architecture or action representation, it uses a pretrained spatial representation as an auxiliary reward signal.

3.4 Manipulation Benchmarks

LIBERO is a simulated robot-manipulation benchmark for lifelong robot learning and knowledge transfer [11]. It provides task suites such as LIBERO-10 and LIBERO-Spatial, and uses rollout success rate as the primary evaluation metric.

The evaluation focuses on LIBERO-10 because it matches the public official checkpoints used in the controlled experiments. Let S denote success rate, N_{success} denote the number of successful rollouts, and N_{total} denote the total number of rollouts. The metric is computed as:

$$S = \frac{N_{\text{success}}}{N_{\text{total}}}. \quad (6)$$

Following the official protocol, a full LIBERO-10 run contains about 50 trials per task.

3.5 Offline Reinforcement Learning

Offline RL trains policies from a fixed dataset without additional environment interaction, which is attractive when online robot rollouts are expensive. IQL avoids direct evaluation of unseen policy actions during policy improvement [12], while CQL regularizes the Q-function to reduce overestimation on out-of-distribution actions [13].

The implementation applies these offline-RL ideas to a large VLA policy and augments the reward with a spatial representation term.

4. Experimental Results

4.1 Experimental Design

The experiments address one controlled question: whether, under limited compute and without online RL interaction, SpaRe can improve an official supervised VLA policy. The study starts from the author-provided OpenVLA-OFT SFT checkpoint and trains two variants under the same offline data and hyperparameters, changing only whether the auxiliary R_2 spatial-alignment reward is enabled.

Two official author-provided Hugging Face checkpoints serve as references:

Table 1: Official checkpoints used as evaluation references.

Reference	Role
Official SFT checkpoint	SFT baseline
Official post-RL checkpoint	online-RL reference

The SFT checkpoint is the primary baseline for the controlled experiment. The post-RL checkpoint is included as a reference for the performance that full online RL can achieve; the constrained offline setting is intended to test whether the reward formulation produces measurable improvement, not to match a fully online RL upper bound.

4.2 Dataset and Training Pipeline

This work uses offline RL because the setting does not assume access to expensive online interaction. The final controlled dataset combines high-quality demonstrations with realistic low-quality behavior and is stored as `libero10_expert_plus_sft_failures_409.jsonl`. This dataset is denoted full409 because it combines the full LIBERO-10 expert transitions with 409 failed SFT rollout episodes.

Table 2: Composition of the final LIBERO-10 offline training dataset.

Source	Transitions	Description
Expert demos	138,090	Full LIBERO-10 expert transitions
SFT failures	26,176	Real failed SFT episodes
Total	164,266	Expert + failure transitions

The 409 failed episodes come from the official SFT checkpoint. They are not treated as behavior-cloning positives; instead, metadata such as `quality`, `reward`, and `rollout_success` allows the value/Q components to distinguish expert behavior from failed behavior.

The training pipeline is:

1. Extract full LIBERO-10 expert transitions.
2. Run the official SFT policy online in LIBERO-10 to collect real failed rollouts.
3. Merge expert and failure transitions into a labeled offline dataset.
4. Train R_1 and $R_1 + \lambda R_2$ variants from the same official SFT checkpoint.
5. Export the adapted partial checkpoints into full VLA checkpoint directories.
6. Evaluate the exported checkpoints through the official SimpleVLA-RL LIBERO rollout pipeline.

Evaluation preserves the official action decoding, rollout logic, success metric, and benchmark tasks.

4.3 Configuration Choices

The primary SFT-baseline offline RL experiment uses:

Table 3: Hyperparameters for the controlled LIBERO-10 offline RL runs.

Hyperparameter	Value
Training steps	500
Batch size	16
Gradient accumulation	2
Learning rate	5×10^{-6}
IQL expectile	0.7
CQL alpha	0.1
Only R_1 : λ_{align}	0.0
$R_1 + \lambda R_2$: λ_{align}	0.2
R_2 bias	0.10
Eval batch size	1 rollout / worker

The 500-step budget, batch size 16, and conservative learning rate were chosen to keep the experiment reproducible within the available compute window while avoiding large destructive updates to the pretrained VLA policy.

The auxiliary-reward coefficient was fixed to $\lambda_{\text{align}} = 0.2$ before evaluation. This value is treated as a conservative design choice rather than as a tuned optimum: it gives the spatial-alignment term sufficient weight to affect the policy update while keeping the task/data reward R_1 as the dominant signal. The R_1 -only model sets $\lambda_{\text{align}} = 0$, so the comparison between the R_1 -only run and the $R_1 + \lambda R_2$ run isolates the effect of adding the auxiliary spatial reward under otherwise matched training conditions. No post-hoc sweep over λ_{align} was performed, preserving the experiment as a controlled one-factor comparison under a limited training budget.

The OpenVLA-OFT policy is large, so training updates only a small subset of modules close to action prediction:

```
language_model.model.layers.30
language_model.model.layers.31
language_model.norm
language_model.lm_head
multi_modal_projector
```

This choice avoids full-parameter fine-tuning.

The experiments were run on a remote AutoDL instance with two RTX PRO 6000 GPUs. LIBERO evaluation uses batch size 1 because each rollout is an interactive environment process.

4.4 LIBERO-10 Evaluation Results

Table 4 reports a reference experiment from the official post-RL checkpoint using a spatial-offline data source. It is not the primary controlled comparison for the final claim, but it provides supporting evidence about the relative value of R_1 alone versus a combined $R_1 + \lambda R_2$ reward:

Table 4: LIBERO-10 evaluation for the post-RL baseline and spatial-data offline variants.

Model / Setting	Success	Success rate
Official SFT	80 / 496	16.13%
Official post-RL	430 / 496	86.69%
R_1 from post-RL + spatial data	355 / 448	79.24%
$R_1 + \lambda R_2$ from post-RL + spatial data	364 / 448	81.25%

This reference result, summarized in Table 4 and visualized in Figure 2, indicates that the combined reward improves over R_1 alone by 9 successes, or approximately 2.0 percentage points. Because it uses a different offline data source from the main LIBERO-10 full409 experiment, it is reported as supporting evidence rather than as the primary controlled comparison.

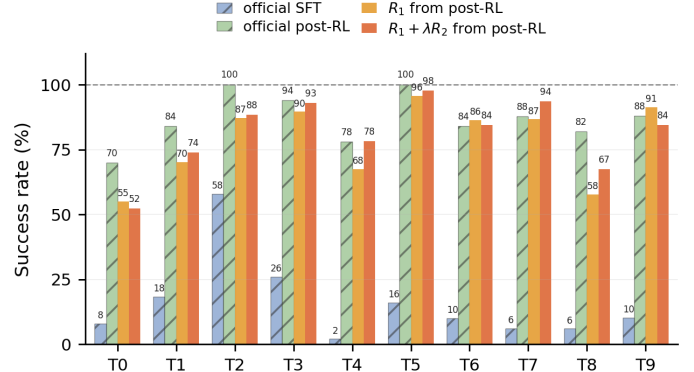


Figure 2: Per-task LIBERO-10 success rates for the official SFT baseline, official post-RL baseline, and post-RL spatial-data offline variants. T0–T9 denote the LIBERO-10 task IDs defined in Table 6.

The main controlled experiment starts from the official SFT checkpoint and trains R_1 and $R_1 + \lambda R_2$ variants on the final LIBERO-10 full409 dataset. Table 5 reports the result:

Table 5: Controlled LIBERO-10 evaluation from the official SFT baseline.

Model / Setting	Success	Success rate
Official SFT	80 / 496	16.13%
R_1 from SFT + full409	86 / 500	17.20%
$R_1 + \lambda R_2$ from SFT + full409	92 / 500	18.40%
Official post-RL	430 / 496	86.69%

The controlled ordering is:

$$\text{Official SFT} < R_1 \text{ from SFT} < R_1 + \lambda R_2 \text{ from SFT.} \quad (7)$$

The R_1 -only variant improves over the official SFT baseline by 1.07 percentage points, and the $R_1 + \lambda R_2$ run improves over SFT by 2.27 points. The absolute gains are modest, but they are consistent with the constrained setting: no online RL, a 500-step training budget, and restricted trainable policy layers.

Figure 3 reports the per-task comparison for the main controlled experiment, complementing the aggregate results in Table 5.

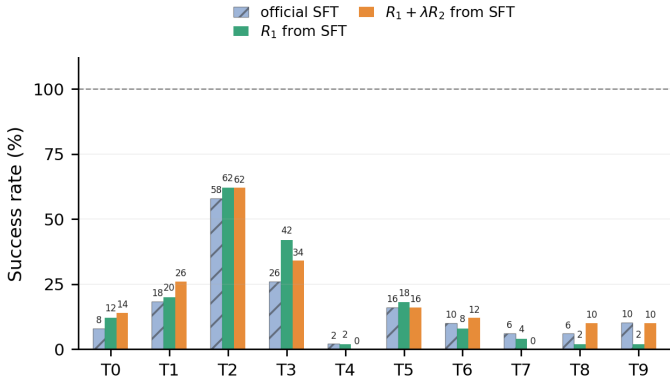


Figure 3: Per-task LIBERO-10 success rates for the official SFT baseline and the controlled full409 offline variants. T0–T9 denote the LIBERO-10 task IDs defined in Table 6.

Together, these evaluations support the central claim of this study: the main controlled experiment yields $SFT < R_1 < R_1 + \lambda R_2$, and the reference experiment yields the same combined-reward-over- R_1 ordering from a higher-performing initialization.

Table 6: LIBERO-10 task mapping used in Figures 2 and 3.

ID	Task description
T0	soup + tomato sauce → basket
T1	cream cheese + butter → basket
T2	turn on stove + put moka pot
T3	black bowl → bottom drawer and close
T4	white mug left plate + yellow/white mug right plate
T5	book → back compartment of caddy
T6	white mug on plate + pudding right of plate
T7	soup + cream cheese → basket
T8	both moka pots → stove
T9	yellow/white mug → microwave and close

5. Conclusion and Future Work

The results provide evidence for a constrained but reproducible spatial-reward direction for VLA reinforcement learning. Without online RL interaction, a short 500-step offline RL run, and limited trainable policy layers, the R_1 -only run improves over the official SFT baseline and the $R_1 + \lambda R_2$ run further improves over R_1 . This supports the hypothesis that auxiliary spatial alignment can provide a measurable positive signal, even when the improvement is modest.

The main limitation is that the final implementation is offline and resource-constrained. The official post-RL checkpoint remains substantially higher-performing because online RL can collect fresh rollouts and directly optimize environment success. The model also updates only a small subset of policy layers.

A primary future direction is to implement the online version of SpaRe by adding the R_2 spatial representation reward to the SimpleVLA-RL rollout-based training loop. With more compute, future work would also unfreeze more policy layers, train longer, and evaluate whether R_2 improves learning speed and stability.

Additional future work includes broader LIBERO evaluation, more expressive spatial encoders, richer spatial reward

terms, and more random seeds.

Author Contributions

All team members contributed to technical discussion, experiment review, and report writing. Primary responsibilities were:

Table 7: Summary of team contributions.

Contribution	Haochen	Yan	Albert	Jiayi
Research direction and method design	✓			
Environment and infrastructure support	✓	✓		
Model training and experiment execution	✓	✓	✓	✓
Evaluation and reproducibility documentation	✓	✓	✓	✓
Report writing and editing	✓	✓	✓	✓

Appendix

A. Reproducibility Package

The public reproduction repository is:

<https://github.com/Leo-Haochen-Liu/spare-lite-libero10-offline-rl>

The repository contains:

Table 8: Contents of the public reproduction repository.

Directory	Purpose
<code>data/</code>	Final JSONL dataset and manifest
<code>spare_lite/</code>	Extraction, rewards, adapters
<code>spare_lite_offline_rl/</code>	IQL-style offline RL
<code>scripts/</code>	Failed-rollout collection
<code>tools/</code>	Export/download/plot utilities
<code>figures/</code>	Pipeline and result figures
<code>analysis/</code>	Parsed rollout metadata
<code>results/</code>	Final logs and summaries
<code>patches/</code>	Minimal runtime patch

The four full exported VLA checkpoints used in the reported comparisons are published on Hugging Face. The first pair supports the primary controlled SFT-baseline full409 experiment; the second pair is the post-RL spatial-data reference pair discussed in Table 4.

Table 9: Published checkpoint artifacts for reproducing the reported evaluations.

Checkpoint	Hugging Face link
R_1 from official SFT + full409	R₁ checkpoint
$R_1 + \lambda R_2$ from official SFT + full409	R₁ + λR_2 checkpoint
R_1 from official post-RL + spatial data	R₁ post-RL spatial checkpoint
$R_1 + \lambda R_2$ from official post-RL + spatial data	R₁ + λR_2 post-RL spatial checkpoint

External downloads required for reproduction include the official SFT checkpoint, the official post-RL checkpoint, SimpleVLA-RL, LIBERO, and the SpatialVLA backend checkpoint. The reproduction procedure is: clone the repository with Git LFS, download the official checkpoints, prepare SimpleVLA-RL and LIBERO, pull or regenerate the final JSONL dataset, train R_1 and $R_1 + \lambda R_2$ variants, export partial checkpoints into full VLA checkpoint directories, and run LIBERO-10 evaluation through SimpleVLA-RL.

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